

The voice that uses your computer for you

"yes?"



Ctrl+Alt+M



A voice agent for Windows, heard by *AssemblyAI*, for people who can't see the screen.
team revived · lablab.ai × AssemblyAI Voice Agent Hackathon · September 2026

“Button, button, *button.*”

That is what a screen reader says when an app doesn't label its buttons. Many don't. Blind users then need a sighted person to do it for them, and older users give up.

~95%

of the top million home pages fail
basic accessibility checks

Source: WebAIM Million, 2024-25.



business value

An app that wasn't
built for you *costs 260
hours a year.*

€4,000

and 260 hours a year per
person, in sighted help and
lost time (our estimate)

338 M

people blind or with
moderate to severe vision
loss

\$411 bn

productivity lost to vision
impairment every year

1 in 6

people will be over 60 by
2030, the second audience

Estimate: 4 h a week of sighted help + 1 h lost, at €15/h, plus missed online prices. Sources: Bourne et al., Lancet Global Health 2021, GBD 2019 (43 M blind + 295 M); Marques et al., eClinicalMedicine 2021; WHO.

business value

Say it. *Iris does it*, and asks first.

You say what you want in plain words. Iris does it in any app on your PC, tells you what happened, and waits for your spoken yes before anything that sends, pays or deletes.

YOU	Put vitamin D in my basket.
IRIS	Done. One bottle of vitamin D3 is in your basket.
YOU	Now check out.
IRIS	<i>Check out and pay for your basket? It costs €32.86.</i>
YOU	No, wait.
IRIS	Okay. I didn't buy anything.

Screen readers wait for developers. *Iris doesn't.*

LAYER	USED WHEN	NEEDS THE APP'S DEVELOPER?
1 · An API	The service has one: Gmail through MCP, web pages through browser shortcuts	No
2 · The tree	Kev reads the controls Windows already sees, in about 230 ms	No
3 · The pixels	Unlabelled icons and custom-drawn apps: Claude looks at the screen like a sighted helper	No

Proof: Telegram Desktop (unlabelled send button), Amazon (checkout), Gmail (API). **10 of 10** unlabelled icons found.

Three errands, *real services.*

ERRAND	WHAT YOU HEAR	WHAT IT PROVES
Gmail	“Did Peter write?” It reads the email, writes your reply, asks “Send it to Peter?”, and asks again when you sound unsure	API first, spoken yes
Telegram	“Tell Peter I'm running late.” The send button has no label; vision finds it	Works where screen readers stop
Amazon	Fills the basket, names the price, and “No, wait” means nothing is bought	Money needs a clear yes

Every clip is a real, continuous take on test accounts; sped-up clips are marked 2x. Video and live page: whom-m0rty.github.io/Iris-voice-agent

originality

Both AssemblyAI paths, *one agent*.

REALTIME STT • THE DEMO

Universal-Streaming

- Speech-to-text over WebSocket, formatted turns
- Pauses merged into one turn: older people pause mid-sentence (0.8–4 s, set by voice)
- Speech-start events let you talk over Iris
- Bring our own brain (Claude) and neural TTS

VOICE AGENT API

One connection

- Speech, turn-taking and voice in one WebSocket
- JSON-Schema tools: do_task, answer_confirmation, stop_task, mute_microphone
- Tools answer at once; progress spoken with reply.create
- Runs as a second backend

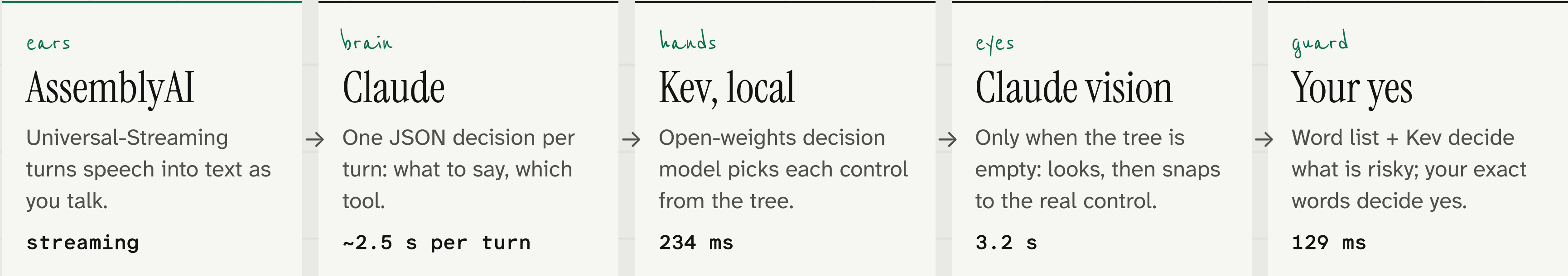
LLM GATEWAY

Swappable brain

- Claude through AssemblyAI's gateway in the stored agent
- Any of the 47 gateway models as the brain, one line in .env
- Qwen tested: 0.85 s per plan

application of technology

Each model does *what it's best at.*



The fast part does almost all the work. The slow part steps in only when it has to. Kev, vision and yes/no latencies: bench/bench_full.py on an RTX 5060 Ti. Brain turn time seen in live runs.

It never acts *on a maybe.*

0

false yeses in 60 spoken answers

0

risky actions let through, in 40 it had never seen

92%

controls picked right by Kev in 25 test steps; misses went to vision, not to a wrong click

10/10

unlabelled icons found by Claude vision

Passwords never reach a model: code types them from Windows Credential Manager. Apps on the do-not-touch list are never read or clicked. Muted, a pending question can only end as a no.

Small behaviours *that change the day.*

“What’s on my screen?”

Where you are, pop-ups first, three things that matter. Photos described by vision.

“I’m lost.”

Says where you are and what box is open. Never closes it without asking.

“Show me how.”

Does the task slowly, explaining each step in plain words. It teaches as it goes.

“Say that again.”

Repeats the last reply word for word, at once.

“Wait longer for me.”

Pause length set by voice, remembered next time.

tick · chime · low tone

Sounds for thinking, done and error. Silence never leaves you guessing.

Free for the person. *Paid by who carries the cost today.*



1 Families

“Mum calls me every day to fix her email.” Remote setup, a daily log, alerts when she's stuck.

€9-12 / month

2 Assistive-tech funders

Rehab agencies, Access to Work, blind associations: budgets exist for approved tools.

€150-400 / seat / year

3 Employers

Internal apps that will never be made accessible become usable, with audit log and do-not-touch list.

€20-40 / seat / month

4 Banks and telcos

Older customers call support for what the app could do. Connect an API; every call avoided is saved cost.

per active user or task

business value

Law, age and cost *all point one way.*

The law. The European Accessibility Act has applied since 28 June 2025 to banking, shopping and telecom.

Age. 1 in 6 people over 60 by 2030 (WHO).

Cost. Voice AI is now fast and cheap enough to do the task itself, where the alternative is a sighted helper's hour.

WHEN	MOVE	PROOF
Q1	Free beta with two blind associations and 20 families	Tasks done without help
Q2	Family plan, case studies	Month-two retention
Q3	One funder's approved list; first employer pilot	First funded seats
Q4	Bank or telco pilot: bills and balances through their API	Support calls avoided

business value

A computer is easy when it *listens* *to you.*

whom-m0rty.github.io/Iris-voice-agent · github.com/Whom-m0rty/Iris-voice-agent

